

Taylor & Francis
Taylor & Francis Group

Nuclear Science and Engineering

ISSN: 0029-5639 (Print) 1943-748X (Online) Journal homepage: www.tandfonline.com/journals/unse20

Calculation of Adjoint-Weighted Kinetics Parameters in OpenMC Using the Iterated Fission Probability Method

Joffrey Dorville , Luke Labrie-Cleary & Paul K. Romano

To cite this article: Joffrey Dorville , Luke Labrie-Cleary & Paul K. Romano (05 Mar 2026): Calculation of Adjoint-Weighted Kinetics Parameters in OpenMC Using the Iterated Fission Probability Method, Nuclear Science and Engineering, DOI: 10.1080/00295639.2026.2613323

To link to this article: https://doi.org/10.1080/00295639.2026.2613323

Published online: 05 Mar 2026.

Submit your article to this journal

Article views: 245

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=unse20

NUCLEAR SCIENCE AND ENGINEERING
© 2026 UChicago Argonne, LLC, Operator of Argonne National Laboratory
DOI: https://doi.org/10.1080/00295639.2026.2613323

Check for updates

Calculation of Adjoint-Weighted Kinetics Parameters in OpenMC Using the Iterated Fission Probability Method

Joffrey Dorville[a], Luke Labrie-Cleary[b], and Paul K. Romano[a]

[a]*Computational Science Division, Argonne National Laboratory, Lemont, IL;* [b]*Dynamic Modelling, Copenhagen Atomics A/S, Kastrup, Denmark*

Received September 30, 2025
Accepted for Publication December 23, 2025

Abstract — *New capabilities to compute adjoint-weighted kinetics parameters have been developed, tested, validated, and made available in OpenMC, an open-source Monte Carlo particle transport code. The effective delayed neutron fraction β_{eff} and the effective generation time Λ_{eff} are calculated using the iterated fission probability (IFP) method. Effective kinetics parameters produced with OpenMC are compared to a prior work performed with a developmental branch of OpenMC and other results from Serpent and MCNP based on several benchmark cases from the ICSBEP (International Criticality Safety Benchmark Evaluation Project). Overall, the results are in good agreement with a maximum difference in β_{eff} of 25 pcm and 2.7% for Λ_{eff}.*

The results of the new IFP implementation are also compared to design data used in theoretical kinetics models for the ARE (Aircraft Reactor Experiment) and the Molten Salt Reactor Experiment (MSRE) when the fuel is not circulating. Reasonable agreement was obtained for both experiments with a maximum difference in β_{eff} of 54 pcm and 13.4% for Λ_{eff}.

This new feature is fully compatible with shared- and distributed-memory parallelism. The impact of the new implementation on code performance was verified to be minimal when IFP was not used with a maximum relative difference of 0.2% in peak memory usage and 0.8% in execution time for the model considered. When using the new IFP feature, the code performance scaled with the number of processes and threads, as expected.

Keywords — *Kinetics parameters, iterated fission probability, OpenMC.*

I. INTRODUCTION

Kinetics parameters play an essential role in the safe operation of nuclear reactors. Having the capability to compute these parameters accurately is important to justify the safety of existing nuclear reactors and for the development of advanced designs. Effective kinetics parameters for a point reactor are derived from transport theory by using the forward and adjoint transport equations [1]. Using the notion of adjoint flux, the effective delayed neutron fraction β_{eff} and the effective generation time Λ_{eff} can be defined as [1]:

$$\beta_{eff} = \frac{\langle \psi^{\dagger}, D_f \psi \rangle}{\langle \psi^{\dagger}, T_f \psi \rangle} , \tag{1}$$

$$\Lambda_{eff} = \frac{\langle \psi^{\dagger}, \frac{1}{v} \psi \rangle}{\langle \psi^{\dagger}, T_f \psi \rangle} , \tag{2}$$

where the notation $\langle x, y \rangle$ represents the integral of the product of x and y over the integration domain, $\psi^{\dagger}$ is the adjoint of the neutron flux ψ, D_f is the delayed fission production term, T_f is the total fission production term, and v is the velocity of neutrons. Because the adjoint flux acts as a weighting function for β_{eff} and Λ_{eff}, these quantities are referred to as adjoint-weighted kinetics parameters. The effective delayed fraction β_{eff} represents the total fraction of delayed neutrons that are

CONTACT Joffrey Dorville jdorville@anl.gov

present in the system at a given time weighted by the adjoint flux [1]. In practice, an instantaneous reactivity insertion in excess of this value will make a nuclear system reach prompt criticality. The effective generation time Λ_{eff} can be interpreted as how long it takes for a neutron to "reproduce itself" by fission [2].

Eqs. (1) and (2) both use the neutron adjoint flux, which is not trivially obtained from standard Monte Carlo (MC) neutron transport simulations. While it is possible to solve for the adjoint flux directly in multigroup MC neutron transport by adjusting the fission and scattering terms [3], this technique is not applicable to continuous-energy MC neutron transport [4,5]. A common approach used to approximate the effective kinetics parameters is to rely on the difference between a reference and a perturbed state of a system (e.g., the "prompt method" [6] to calculate β_{eff} and the "$1/v$ absorber method" [7] to calculate Λ_{eff}). The main drawback of this approach is that it requires two full calculations of the same system. Another approach that has been widely adopted among MC neutron transport solvers, including MCNP [5], Serpent [2], and TRIPOLI [8], is the iterated fission probability (IFP) method [9]. The foundational idea behind this method is that the number of fission neutrons descending from an initial neutron at a given phase space is proportional to the adjoint flux at this phase space provided that the neutron flux has already converged to its fundamental mode and that the number of generations N_{gen} between the initial neutron and its descendants is sufficiently large [10]. Tracking the population of descendants of a given neutron is possible in continuous-energy MC calculations by storing additional data during iterations, and thus, can be used to evaluate β_{eff} and Λ_{eff}. The only unknown parameter needed to use IFP is the number of generations N_{gen}, which must be large enough to converge the descendant population while limiting the memory burden and avoiding an increase in statistical uncertainty observed when N_{gen} is large [8]. Generally, this parameter is problem dependent and difficult to predict [5]. A number of generations between 10 and 20 is assumed to be sufficient for most cases [8]. A specific case of IFP where only the next generation is considered is known as the next fission probability (NFP) method. More recent work from Peng et al. [11] used a developmental branch of OpenMC [12] to compute adjoint-weighted kinetics parameters from the CLUTCH (contributon-linked eigenvalue sensitivity/uncertainty estimation via tracklength importance characterization) method described in Ref. [3]. This prior work, which also included an implementation of IFP and NFP, was never merged into the main branch of OpenMC as it needed further development to enable its integration.

In this paper, a new implementation of IFP in OpenMC based on the Serpent approach [2] is presented. This implementation has been coded from the ground up and merged to the develop branch of the main OpenMC repository (commit 47ca2916) via a pull request[a]. The IFP implementation was later corrected and merged to the develop branch (commit 767db7e6a) via another pull request[b]. The main focus of this work is to guarantee that the new implementation has minimal impact on performance when IFP is not used and that using multiple processes when IFP is used still improves the performance of the code. Several benchmark cases from the International Criticality Safety Benchmark Evaluation Project (ICSBEP) [13] were used to validate the results of the new IFP implementation in OpenMC against the results from Serpent and MCNP reported in Ref. [2] and the results from Peng et al [11]. Additionally, results from the new implementation are compared to design data used in theoretical kinetics models for the Aircraft Reactor Experiment (ARE) and the Molten Salt Reactor Experiment (MSRE). Particular efforts have been made to minimize the impact of this IFP implementation on memory and execution time. A performance analysis is provided to assess the impact on peak memory usage and execution time.

II. METHODS

Proportionality between the adjoint flux at a given location in phase space and the number of descendants in a future generation from a common ancestor at the same location can be demonstrated using a balance equation and proving that the number of descendants is a solution to the adjoint transport equation [10]. The underlying assumption of this demonstration is that the neutron flux in the generation that saw the emission of the common ancestor has converged to its fundamental mode and that the number of generations between the ancestor and the descendants is large enough to see the convergence of the number of descendants [10]. Given the form of Eqs. (1) and (2), having access to a quantity that is proportional to the adjoint flux, namely, the number of descendants in a future generation under proper conditions, is a sufficient condition to solve for the effective

[a]Pull request #3133 (https://github.com/openmc-dev/openmc/pull/3133).
[b]Pull request #3580 (https://github.com/openmc-dev/openmc/pull/3580).

kinetics parameters, as proportionality coefficients cancel out.

The normalized adjoint-weighted delayed fission production term, corresponding to the numerator of Eq. (1) divided by the effective multiplication factor k_{eff}, can therefore be seen as proportional to the sum of weights W_{delayed} from all neutrons emitted in a generation g that descend exclusively from delayed neutrons emitted in generation o, which occurs N_{gen} generations before g [2]. The normalization of the adjoint-weighted delayed fission production term by k_{eff} results from the power iteration method used in MC simulations to guarantee a stable population over generations. The quantity W_{delayed} can be estimated by tallying the score $T_{1,g}$ [Eq. (3)],

$$\frac{1}{k_{eff}}\ \psi^{\dagger}, D_f\psi\ \propto W_{\text{delayed}} \approx T_{1,g} = \sum_{n\in F_g} w_n \delta_{a_n}\,, \tag{3}$$

where n is any neutron emitted by fission in generation g causing a new fission event leading to the creation of new fission neutrons in generation $g+1$. For generation g, these neutrons n form the set F_g. The symbol w_n corresponds to the particle weight of the neutron n. The symbol a_n corresponds to the ancestor of a neutron n that was emitted during generation o, and δ_{a_n} is equal to 1 if the ancestor a_n was a delayed neutron or 0 if it was a prompt neutron. Because the population of descendants is not known when the ancestor is emitted in generation o in a MC simulation, the evaluation of adjoint-weighted contributions with Eqs. (1) and (2) can only be performed N_{gen} generations later, when the descendants are emitted. In practice, ancestors' contributions to a specific tally need to be passed from each generation to the next by keeping the contribution and the ancestor/descendant relation in memory. A first algorithm developed in MCNP [5] consists of separating the total number of active generations into blocks composed of an origin generation, multiple latent generations, and a final generation. The descendant population of multiple ancestors emitted in the origin generation is tracked until the final generation is reached.

During the final generation, ancestors contributions [e.g., δ_{a_n} in Eq. (3)] are weighted by their descendants' weights and tallied. A second algorithm developed in Serpent [2] and adopted in TRIPOLI [8] and the present implementation in OpenMC considers every active generation as an origin, latent, and final generation instead of relying on the block logic. While this approach requires more memory as all the particles are potentially contributing to the tallies, it is easier to implement [2] and increases the number of realizations for the associated tallies [8].

Similarly, the normalized adjoint-weighted total neutron fission production term corresponding to the denominator of Eqs. (1) and (2) divided by k_{eff} is proportional to the sum of weights W_{total} from all neutrons emitted in g [2] and can be tallied using the score $T_{2,g}$ [Eq. (4)],

$$\frac{1}{k_{eff}}\ \psi^{\dagger}, T_f\psi\ \propto W_{\text{total}} \approx T_{2,g} = \sum_{n\in F_g} w_n. \tag{4}$$

The numerator of the effective generation time Λ_{eff} is directly proportional to the sum of weights W_{lifetime} from all neutrons emitted in generation g multiplied by the lifetime of their respective ancestor that was emitted in generation o [2]. The quantity W_{lifetime} can be tallied using the score $T_{3,g}$ [Eq. (5)],

$$\psi^{\dagger}, \frac{1}{v}\psi\ \propto W_{\text{lifetime}} \approx T_{3,g} = \sum_{n\in F_g} w_n l_{a_n}\,, \tag{5}$$

where a_n (emitted in o) is the ancestor of the neutron n (emitted in g). In OpenMC, the lifetime l_{a_n} of the ancestor a_n is obtained by tracking the time between its birth and death. When secondary particles are emitted from reactions such as (n,2n), the secondary particle is considered born when this particular event occurs. The effective delayed neutron fraction and effective generation time can then be estimated by tallying the previously defined scores ($T_{1,g}$, $T_{2,g}$, and $T_{3,g}$) for each active generation g using IFP tallies (T_1, T_2, and T_3, respectively) until all N active generations are simulated [Eqs. (6) and (7)],

$$\beta_{eff} \approx \frac{\sum_{g=1}^{N} T_{1,g}}{\sum_{g=1}^{N} T_{2,g}} = \frac{T_1}{T_2}\,, \tag{6}$$

$$\Lambda_{eff} \approx \frac{1}{k_{eff}} \frac{\sum_{g=1}^{N} T_{3,g}}{\sum_{g=1}^{N} T_{2,g}} = \frac{1}{k_{eff}} \frac{T_3}{T_2}\,. \tag{7}$$

Because the quantities involved in Eqs. (6) and (7) are tallied independently, their respective standard deviations are also evaluated independently, following the strategy that is normally used for regular tally estimators in OpenMC. The standard deviations $\sigma_{\beta_{eff}}$ and $\sigma_{\Lambda_{eff}}$ related to the estimators of the adjoint-weighted delayed neutron

fraction β_{eff} and the adjoint-weighted generation time Λ_{eff}, respectively, are calculated from Eqs. (8) and (9),

$$\sigma_{\beta_{eff}} = \beta_{eff} \times \sqrt{\left(\frac{\sigma_{T_1}}{T_1}\right)^2 + \left(\frac{\sigma_{T_2}}{T_2}\right)^2}\ , \tag{8}$$

$$\sigma_{\Lambda_{eff}} = \Lambda_{\text{eff}} \times \sqrt{\left(\frac{\sigma_{T_3}}{T_3}\right)^2 + \left(\frac{\sigma_{T_2}}{T_2}\right)^2 + \left(\frac{\sigma_{k_{eff}}}{k_{eff}}\right)^2}\ , \tag{9}$$

where the σ_i symbol corresponds to the 1σ standard deviation associated with the given tallied quantity i. While the tallies involved in the calculation of the effective kinetics parameters are obtained from the same simulation, the uncertainties from Eqs. (8) and (9) are calculated by assuming that these tallies are uncorrelated variables. The absence of covariance data for this specific application is the essential reason for this simplification.

III. IMPLEMENTATION IN OPENMC

To obtain the effective kinetics parameters as defined in Eqs. (6) and (7), a record of the precursor group (setting 0 for prompt neutrons by convention) and lifetime of a given set of ancestors has to be kept for each neutron transported during the simulation. The number of consecutive neutron ancestors to consider in the IFP method is referred to as the number of generations N_{gen}. The optimal value of N_{gen} is model dependent and is generally found empirically by monitoring the convergence of kinetics parameters for different values of N_{gen}. N_{gen}, therefore, has been added as a user-defined parameter in the settings of OpenMC. It is considered constant during the entire simulation, and its default value is set to 10. N_{gen} can be set by the user via the Python application programming interface (API) from the following commands:

```
settings = openmc.Settings()
settings.ifp_n_generation = 15
```

Because the number of generation N_{gen} is determined at run time, ancestors' lifetimes and precursor groups are stored in dynamic data containers (vectors). The memory overhead associated with dynamic data containers allocated at run time can have a nonnegligible impact on the total memory footprint, especially when data containers are present in a significant number and even if these containers are empty (i.e., when the IFP feature is not requested by the user). To ensure the memory impact of the IFP feature on OpenMC is negligible when it is not requested, IFP-related data (precursor group numbers and lifetimes of ancestors) are not stored as particle attributes, but rather in four global data banks (vectors of vectors). Two of these banks are dedicated to storing vectors of ancestor precursor group numbers for every neutron, and the other two contain vectors of ancestor lifetimes. The power iteration process used in OpenMC requires two separate banks per type of data type. One bank (called the source bank) is used to store the information relative to the source particles of a given generation, while the second bank (called the fission bank) holds the information relative to the fission sites that result from the transport of particles during this generation and that will be used as sources in the following generation. IFP data sets relative to a given source particle are stored in IFP source banks using the same index that was used to store the source particle in the source bank. Similarly, IFP data sets stored in the IFP fission banks use the same index that was used to store the fission site in the fission bank. This approach guarantees that the IFP implementation in OpenMC is thread-safe and can work with multiple processes. While using the same data banks for both data types with structs containing one integer and one floating point value (for the precursor groups and the lifetimes, respectively) was initially seen as a more effective way to transfer memory between processes during parallel execution, the extra padding added to the struct for memory alignment was deemed unacceptable. Having two separate data banks also offers the added benefit of being able to calculate β_{eff} and Λ_{eff} independently.

Fig. 1 illustrates the integration of the IFP feature in the surface tracking algorithm of OpenMC using a simplified representation. Modifications performed on OpenMC to implement the IFP feature are represented in blue boxes. During the simulation, a particle is created by sampling available particles from the source bank. The first step of the transport algorithm is to identify the distance to the next collision. The lifetime of the particle is updated and stored as a particle attribute. If no surfaces are crossed before the collision occurs, the type of collision is sampled. If a fission is sampled, new fission sites are created and stored in the fission bank. The calculated lifetime of the particle that generated the fission and its precursor group are stored in their

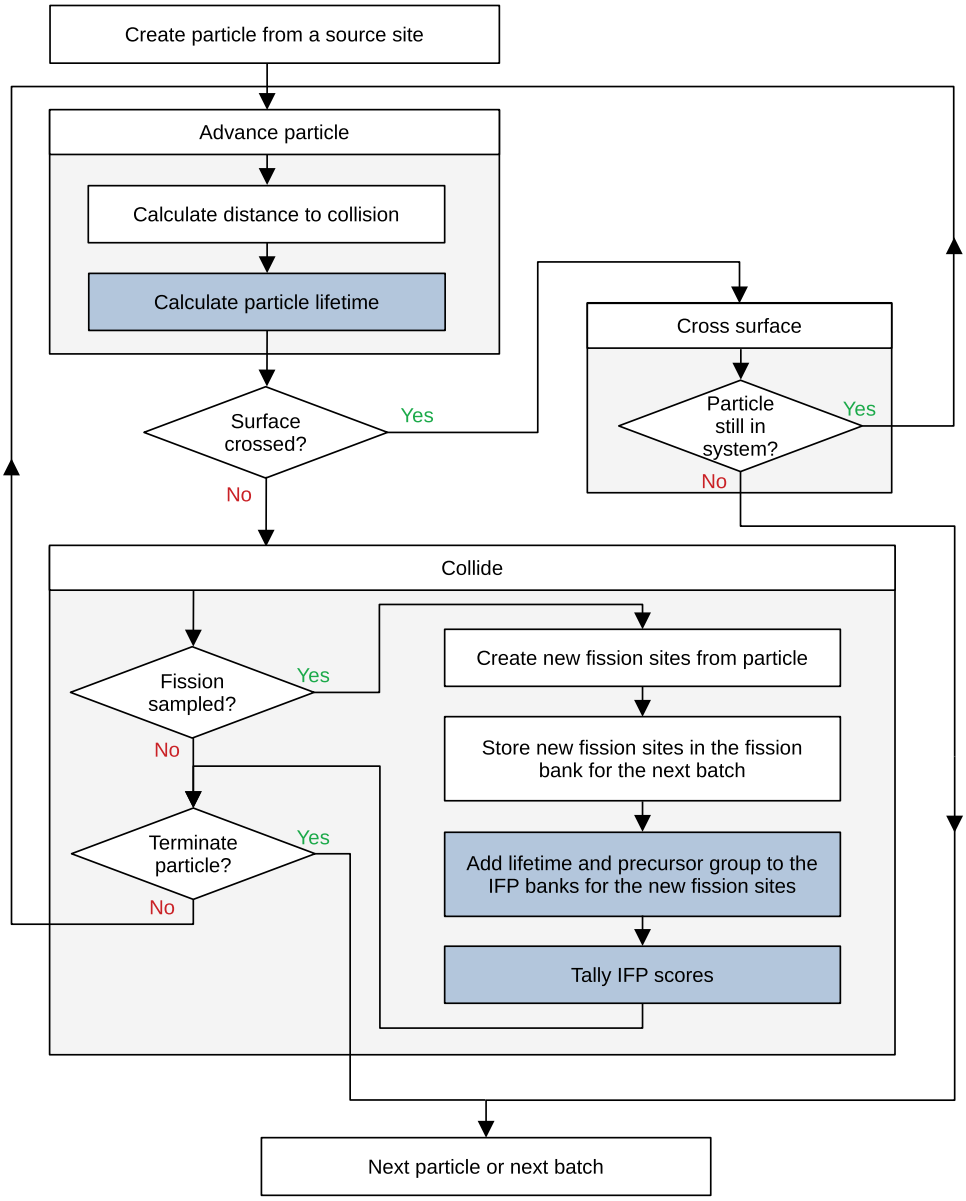


Fig. 1. Integration of the IFP feature in the surface tracking algorithm of OpenMC.

respective fission banks for each fission site using the same index as the fission site index. IFP scores are then tallied from the IFP data stored in the source banks. If the particle needs to be terminated, the next particle or next batch (generation) can be processed.

Three tallies, which have been specifically implemented to obtain effective kinetics parameters from the IFP method, are named ifp-time-numerator, ifp-beta-numerator, and ifp-denominator, corresponding to the symbols T_3, T_1, and T_2 in Eqs. (6) and (7), respectively. All three tallies use a collision estimator and can be set via the Python API of OpenMC from the following commands:

```
tally = openmc.Tally(name="ifp")
tally.scores = [
    "ifp-time-numerator",
    "ifp-beta-numerator",
    "ifp-denominator"
]
```

Calculating the effective delayed neutron fraction and effective generation time is done by retrieving the corresponding tally values and k_{eff} from the statepoint

file once the simulation is complete and applying Eqs. (6) and (7).

IV. RESULTS AND VALIDATION

All calculations performed in this section were run on a system with a dual-socket Intel Xeon Platinum 8260 processor with 24 physical cores per processor, each of which had two hardware threads per core for a total of 96 hardware threads. OpenMC was configured to use the latest commit 767db7e6a (including both pull requests).

IV.A. ICSBEP Benchmarks

Table 1 summarizes the ICSBEP benchmark cases selected for computational comparison. The results using Serpent 2.1.15 and MCNP5 v1.60 were obtained from Ref. [2]. The kinetics parameters produced with OpenMC were generated for two different nuclear data libraries, JEFF-3.1.1 [14] and ENDF/B-VII.1 [15], to allow for direct comparison with the results reported in the Serpent analysis [2] and in the work from Peng et al. [11], respectively. Nuclear data based on JEFF-3.1.1 were generated in the OpenMC HDF5 format from the original ENDF files using the openmc.data module of OpenMC and NJOY 2016.74 [16]. For all cases in this section, OpenMC was run using 20 000 particles per batch, 2000 active batches, 100 inactive batches, and to be consistent with the Serpent results in Ref. [2], 15 generations for IFP.

Multiplication factors k_{eff}, effective generation times Λ_{eff}, and effective delayed neutron fractions β_{eff} obtained for the new IFP implementation of OpenMC using JEFF-3.1.1 are presented in Tables 2–4, respectively, along with the results from Serpent and MCNP [2]. The symbols D1 and D2 represent the absolute difference between OpenMC and Serpent (D1) and OpenMC and MCNP (D2), respectively, for the variable of interest. Similarly, the symbols RD1 and RD2 represent the absolute relative difference between OpenMC and Serpent (RD1) and OpenMC and MCNP (RD2), respectively, using the Serpent or MCNP value in the denominator. Overall, the results generated with OpenMC were close to the ones reported for Serpent and MCNP with a maximum difference in Λ_{eff} of 2.2% and 25 pcm for β_{eff}. Some cases showed larger differences in k_{eff} (zpr-u9, zpr-heu, stacy-029, stacy-030, and stacy-033), which might be due to differences in the models used in the respective studies. MC calculations can be very sensitive to material definitions, which may be slightly different in two models due to how densities are normalized. Such discrepancies in the material definition could lead to differences in k_{eff} on the order of several dozens of pcm, and could

TABLE 1

Summary of Selected Cases from the ICSBEP Benchmark [13]

Case	ICSBEP Identifier	Description
godiva	HEU-MET-FAST-001	Sphere of uranium
jezebel[a]	PU-MET-FAST-001	Sphere of plutonium
flattop23	U233-MET-FAST-006	Sphere of ^{233}U reflected by a shell of uranium
topsy[b]	HEU-MET-FAST-028	Sphere of uranium reflected by a shell of uranium
popsy	PU-MET-FAST-006	Sphere of plutonium reflected by a shell of uranium
skidoo	U233-MET-FAST-001	Sphere of ^{233}U
stacy-029	LEU-SOL-THERM-004 (run 29)	Uranyl nitrate solution reflected by water
stacy-030	LEU-SOL-THERM-007 (run 30)	Uranyl nitrate solution
stacy-033	LEU-SOL-THERM-004 (run 33)	Uranyl nitrate solution reflected by water
stacy-046	LEU-SOL-THERM-004 (run 46)	Uranyl nitrate solution reflected by water
bigten	IEU-MET-FAST-007	Mixed uranium metal core reflected by ^{238}U
zpr-heu	HEU-MET-INTER-001	^{235}U cylinder reflected by stainless steel
zpr-mox	MIX-MET-FAST-011	Plutonium and uranium cylinder reflected by graphite
zpr-pu	PU-MET-INTER-002	Plutonium cyclinder reflected by stainless steel and iron
zpr-u9	IEU-MET-FAST-010	^{235}U cylinder reflected by depleted uranium
stacy-125	LEU-SOL-THERM-016 (run 125)	Uranyl nitrate solution reflected by water
stacy-215	LEU-SOL-THERM-021 (run 215)	Uranyl nitrate solution

[a]Uses the original sphere radius available before the 2021 update of the ICSBEP Handbook.
[b]Corresponds to an experiment realized with the flattop system.

TABLE 2

k_{eff} Values for OpenMC, Serpent [2], and MCNP [2] (JEFF-3.1.1 [14])

Case	OpenMC	Serpent [2]	MCNP [2]	D1 (pcm)	D2 (pcm)
godiva	0.99632 ± 0.00010	0.99645 ± 0.00006	0.99638 ± 0.00004	13	6
jezebel	1.00002 ± 0.00010	1.00002 ± 0.00007	1.00013 ± 0.00004	0	11
flattop23	1.00606 ± 0.00012	1.00618 ± 0.00008	1.00604 ± 0.00005	12	2
topsy	1.00166 ± 0.00012	1.00167 ± 0.00007	1.00179 ± 0.00005	1	13
popsy	1.00308 ± 0.00012	1.00277 ± 0.00008	1.00290 ± 0.00005	31	18
skidoo	1.00424 ± 0.00010	1.00411 ± 0.00007	1.00426 ± 0.00004	13	2
stacy-029	1.00188 ± 0.00015	1.00144 ± 0.00006	1.00166 ± 0.00005	44	22
stacy-030	0.99803 ± 0.00014	0.99745 ± 0.00005	0.99752 ± 0.00005	58	51
stacy-033	0.99997 ± 0.00014	0.99939 ± 0.00005	0.99943 ± 0.00004	58	54
stacy-046	1.00187 ± 0.00014	1.00163 ± 0.00005	1.00172 ± 0.00004	24	15
bigten	0.99767 ± 0.00011	0.99774 ± 0.00005	0.99781 ± 0.00004	7	14
zpr-heu	1.01616 ± 0.00013	1.01691 ± 0.00006	1.01697 ± 0.00005	75	81
zpr-mox	0.99061 ± 0.00013	0.99049 ± 0.00007	0.99046 ± 0.00005	12	15
zpr-pu	1.01919 ± 0.00014	1.01891 ± 0.00007	1.01880 ± 0.00005	28	39
zpr-u9	0.99134 ± 0.00011	0.98997 ± 0.00005	0.990010 ± 0.000030	137	133
stacy-125	1.00504 ± 0.00015	1.00468 ± 0.00006	1.00484 ± 0.00005	36	20
stacy-215	0.99812 ± 0.00014	0.99795 ± 0.00004	0.99808 ± 0.00004	17	4

TABLE 3

Λ_{eff} Values for OpenMC, Serpent [2], and MCNP [2] (JEFF-3.1.1 [14])

Case	OpenMC	Serpent [2]	MCNP [2]	Unit	RD1 (%)	RD2 (%)
godiva	5.712 ± 0.008	5.711 ± 0.004	5.714 ± 0.004	10^{-9} s	0.0	0.0
jezebel	2.812 ± 0.004	2.8060 ± 0.0020	2.8090 ± 0.0020	10^{-9} s	0.2	0.1
flattop23	1.2488 ± 0.0035	1.2430 ± 0.0020	1.2450 ± 0.0020	10^{-8} s	0.5	0.3
topsy	1.746 ± 0.004	1.7430 ± 0.0020	1.7400 ± 0.0020	10^{-8} s	0.2	0.3
popsy	1.294 ± 0.004	1.2950 ± 0.0020	1.2990 ± 0.0020	10^{-8} s	0.1	0.4
skidoo	2.6526 ± 0.0035	2.6520 ± 0.0020	2.6490 ± 0.0020	10^{-9} s	0.0	0.1
stacy-029	5.954 ± 0.005	5.9560 ± 0.0020	5.9640 ± 0.0030	10^{-5} s	0.0	0.2
stacy-030	5.858 ± 0.005	5.8570 ± 0.0020	5.8550 ± 0.0030	10^{-5} s	0.0	0.1
stacy-033	6.252 ± 0.005	6.2510 ± 0.0030	6.2710 ± 0.0030	10^{-5} s	0.0	0.3
stacy-046	6.718 ± 0.005	6.7180 ± 0.0030	6.7270 ± 0.0030	10^{-5} s	0.0	0.1
bigten	6.203 ± 0.009	6.147 ± 0.004	6.174 ± 0.004	10^{-8} s	0.9	0.5
zpr-heu	9.832 ± 0.020	9.814 ± 0.008	9.810 ± 0.009	10^{-7} s	0.2	0.2
zpr-mox	5.48 ± 0.05	5.580 ± 0.027	5.603 ± 0.026	10^{-7} s	1.8	2.2
zpr-pu	1.2771 ± 0.0021	1.2760 ± 0.0010	1.2750 ± 0.0010	10^{-6} s	0.1	0.2
zpr-u9	8.757 ± 0.012	8.775 ± 0.006	8.819 ± 0.005	10^{-8} s	0.2	0.7
stacy-125	4.890 ± 0.004	4.8800 ± 0.0020	4.8830 ± 0.0020	10^{-5} s	0.2	0.1
stacy-215	6.682 ± 0.005	6.6800 ± 0.0030	6.6840 ± 0.0030	10^{-5} s	0.0	0.0

impact the kinetics parameters. Note that the differences in k_{eff} for the stacy cases (stacy-029 and stacy-033) may also be due to differences in the processing of the thermal scattering data.

The effective generation times Λ_{eff} and effective delayed fractions β_{eff} obtained for the IFP implementation of OpenMC using ENDF/B-VII.1 are presented in Tables 5 and 6, respectively, along with the results from Peng at al [11]. The symbol RD3 corresponds to the absolute value of the relative difference between the OpenMC and results from Peng et al. [11] using the latter as the reference in the denominator. The results generated with OpenMC were generally close to the results from Peng et al. [11] with the exception of the

TABLE 4

β_{eff} Values in pcm for OpenMC, Serpent [2], and MCNP [2] (JEFF-3.1.1 [14])

Case	OpenMC	Serpent 2 [2]	MCNP5 [2]	D1	D2
godiva	640 ± 7	644 ± 3	647 ± 3	4	7
jezebel	190 ± 3	189 ± 2	191 ± 2	1	1
flattop23	384 ± 5	378 ± 3	378 ± 3	6	6
topsy	695 ± 7	693 ± 3	685 ± 3	2	10
popsy	285 ± 5	285 ± 2	286 ± 2	0	1
skidoo	288 ± 4	290 ± 2	288 ± 2	2	0
stacy-029	752 ± 6	752 ± 3	747 ± 4	0	5
stacy-030	753 ± 6	752 ± 3	763 ± 4	1	10
stacy-033	747 ± 6	741 ± 3	739 ± 4	6	8
stacy-046	726 ± 6	733 ± 3	741 ± 4	7	15
bigten	756 ± 7	731 ± 3	736 ± 3	25	20
zpr-heu	684 ± 7	684 ± 3	687 ± 4	0	3
zpr-mox	362 ± 5	360 ± 2	359 ± 3	2	3
zpr-pu	234 ± 4	229 ± 2	233 ± 2	5	1
zpr-u9	725 ± 7	734 ± 3	733 ± 3	9	8
stacy-125	758 ± 6	764 ± 3	764 ± 5	6	6
stacy-215	734 ± 6	740 ± 3	735 ± 4	6	1

TABLE 5

Λ_{eff} Values for OpenMC and Peng et al. [11] (ENDF/B-VII.1 [15])

Case	OpenMC	Peng et al. [11]	Unit	RD3 (%)
godiva	5.695 ± 0.008	5.691 ± 0.005	10^{-9} s	0.1
jezebel	2.865 ± 0.004	2.8750 ± 0.0020	10^{-9} s	0.3
flattop23	1.263 ± 0.004	1.2650 ± 0.0020	10^{-8} s	0.2
topsy	1.745 ± 0.004	1.7530 ± 0.0020	10^{-8} s	0.4
popsy	1.331 ± 0.004	0.9270 ± 0.0020	10^{-8} s	43.6
skidoo	2.723 ± 0.004	2.7180 ± 0.0020	10^{-9} s	0.2
stacy-029	5.964 ± 0.005	5.8070 ± 0.0030	10^{-5} s	2.7
stacy-030	5.854 ± 0.005	5.8320 ± 0.0030	10^{-5} s	0.4
stacy-033	6.262 ± 0.005	6.1050 ± 0.0030	10^{-5} s	2.6
stacy-046	6.715 ± 0.005	6.5950 ± 0.0030	10^{-5} s	1.8

popsy case, where the relative difference reported for Λ_{eff} was 43.6%. Because the same case gives similar results between Serpent, MCNP, and the new IFP implementation in OpenMC, the authors attribute this major outlier to a potential typographical error in the results reported in Ref. [11] for the Λ_{eff} value of the popsy case. Additionally, the relative differences in Λ_{eff} observed for the water-reflected stacy cases were relatively high (2.7%, 2.6%, and 1.8%), which might indicate a potential difference in the treatment of thermal scattering data for water. Excluding the popsy case, the maximum relative difference for Λ_{eff} was 2.7% and 10 pcm for β_{eff}.

IV.B. Reactor Models

As a promising next-generation design concept, molten salt reactors (MSRs) remain under active research and development today. MSR research began at Oak Ridge National Laboratory (ORNL) in the 1950s. Data from these experiments can be used to validate the IFP method for future use in MSR designs. The MSRE was a 10-MW MSR constructed at ORNL [17] that operated from 1965 to 1969. The ARE was a 2.5-MW MSR built at ORNL as a prototype for an aircraft reactor that operated for 2 weeks; ARE preceded the MSRE by 10 years [18]. Effective kinetics

TABLE 6

β_{eff} Values in pcm for OpenMC and Peng et al. [11] (ENDF/B-VII.1 [15])

Case	OpenMC	Peng et al. [11]	D3
godiva	654 ± 7	651 ± 4	3
jezebel	182 ± 3	183 ± 2	1
flattop23	360 ± 5	367 ± 3	7
topsy	688 ± 7	692 ± 4	4
popsy	281 ± 5	277 ± 2	4
skidoo	284 ± 4	289 ± 2	5
stacy-029	723 ± 6	728 ± 3	5
stacy-030	737 ± 6	733 ± 3	4
stacy-033	733 ± 6	723 ± 3	10
stacy-046	714 ± 6	714 ± 3	0

parameters were not directly measured during these experiments. However, theoretical models were designed prior to the experiments to predict several aspect of reactor operations, including stability and kinetics analysis [17], and prepare operational procedures [18]. In this section, the computational results obtained with OpenMC are compared to the design data used in the theoretical models reported for the ARE and MSRE [17,18].

The OpenMC models used in this paper for the ARE and MSRE are based on openly available computer-aided design (CAD) models developed by Copenhagen Atomics[c]. The CAD files were meshed using the CAD-to-OpenMC tool [19]. For the calculations in this section, OpenMC was configured to use DAGMC [20] with a ray tracer built on Embree [21] provided by the double-down [22] wrapper, instead of the legacy ray tracer based on MOAB [23], to obtain better scalability performance, as shown in Ref. [24]. Both reactors were modeled with no fuel circulation, as this feature is not available in OpenMC yet. The models were run with the same simulation parameters as the ones used in Sec. IV.A: 20 000 particles per batch, 2000 active batches, 100 inactive batches, and 15 generations for IFP.

Tables 7 and 8 show the results of the IFP implementation in OpenMC using ENDF/B-VIII.0 [25] for the generation time and effective delayed neutron fraction, respectively, for the ARE and MSRE compared to the design data. Estimates of the effective delayed neutron fractions using the IFP implementation showed fair agreement with the reported values in both cases, with a maximum difference of 54 pcm.

[c]ARE: https://github.com/openmsr/are and MSRE: https://github.com/openmsr/msre.

TABLE 7

Λ_{eff} Values from Design Data and OpenMC (ENDF/B-VIII.0 [25])

Case	OpenMC (μs)	Design (μs)	Relative Difference (%)
ARE	128.28 ± 0.14	–	–
MSRE	272.16 ± 0.23	[a]2.4 x 10^2	13.4

[a]p. 9 of Ref. [17].

TABLE 8

β_{eff} Values (pcm) from Design Data and OpenMC (ENDF/B-VIII.0 [25])

Case	OpenMC	Design	Difference
ARE	676 ± 6	[a]730	54
MSRE	697 ± 6	[b]666	31

[a]Fig. D2, p.146, in Ref. [18].
[b]p. 9 in Ref. [17].

The effective generation time was not available in the design and experimental reports for the ARE. For the MSRE, the reported value had a relative difference of 13.4% to the computational value calculated with OpenMC.

V. PERFORMANCE EVALUATION

Performance of the new implementation was evaluated by measuring the execution time and memory usage using the stacy-029 (LEU-SOL-THERM-004) model, run 29, from the ICSBEP [13]. All calculations performed in this section were run on the same hardware as in Sec. IV. OpenMC was run using 400 000 particles per batch for 20 inactive plus 10 active batches and used nuclear data from ENDF/B-VIII.0 [25]. The execution times were obtained by taking the average time of five consecutive OpenMC runs. This approach ensured that external factors like job scheduling had a limited impact on the evaluation. Memory usage was obtained by taking the average of the peak memory usage from five consecutive OpenMC runs. Individual memory usage values for OpenMC processes were measured every 0.5 ms by taking the proportional set size reported by the psutil Python package. Peak memory usage values for each process were obtained by taking the maximum value over all

time steps. The peak memory usage for a run was then obtained by summing the peak memory usage values for every process.

V.A. Impact of the New IFP Implementation

The new IFP implementation introduced data structures in both the message passing interface (MPI) and non-MPI parts of the source code. To ensure that the performance was not impacted regardless of whether the executable was compiled with or without MPI, both configurations were evaluated. Tables 9 and 10 show a comparison of the performance metrics between a reference state and a new state containing the IFP implementation. The reference state corresponded to the state of the OpenMC source code before IFP was implemented (commit c1a4d43da in the main OpenMC repository). Because IFP was fully implemented from two pull requests, OpenMC developments that happened between these two pull requests should not be taken into account to allow for a fair comparison with the reference state. This is why the new state considered here was built from a custom branch based on the first time the IFP feature was merged into the develop branch (commit 47ca2916 in the main OpenMC repository) and the correction (commit 767db7e6a) was applied directly after. All runs in this section used 24 threads and a single process. As expected, the implementation had no significant impact on code performance when the IFP method was not used.

The impact of using the IFP method was evaluated by running the model with IFP on and off. The corresponding results are reported in Table 11. Compared to a reference calculation, using IFP in the same conditions with the same model for 15 generations significantly increased both the execution time (42.1%) and memory usage (45.1%). This result was expected, as IFP ultimately requires storing additional attributes for each particle simulated (to store the precursor groups and lifetimes of their ancestors).

V.B. Parallel Performance and Scalability

For this analysis, OpenMC was configured to use the new state as defined in Sec. V.A. Fig. 2 represents

TABLE 9

Peak Memory Usage of the New Implementation When IFP Was Not Used

MPI Enabled	Reference (GB)	New (GB)	Difference (MB)	Relative Difference (%)
No	0.760 ± 0.002	0.761 ± 0.001	1	0.2
Yes	0.794 ± 0.004	0.792 ± 0.004	2	0.2

TABLE 10

Execution Time of the New Implementation When IFP Was Not Used

MPI Enabled	Reference (s)	New (s)	Difference (s)	Relative Difference (%)
No	54.45 ± 0.10	54.88 ± 0.07	0.43	0.8
Yes	56.27 ± 0.20	56.34 ± 0.27	0.07	0.1

TABLE 11

Impact of Using IFP on OpenMC Performance

Metric	IFP Off	IFP On	Difference	Relative Difference
Peak memory usage	0.792 ± 0.004 GB	1.149 ± 0.000 GB	357 MB	45.1%
Execution time	56.34 ± 0.27 s	80.1 ± 0.6 s	23.72 s	42.1%

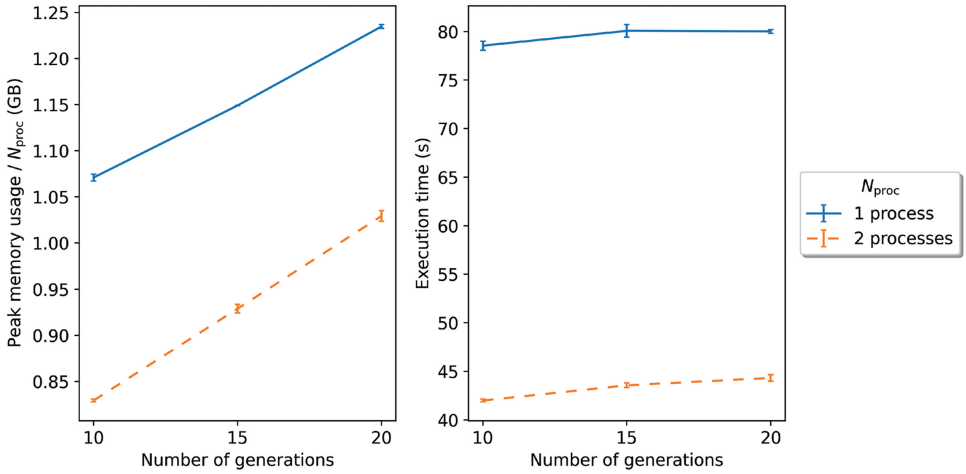


Fig. 2. Peak memory usage per process and execution time as a function of the number of IFP generations and processes for parallel calculations.

the peak memory usage divided by the number of processes and the execution time obtained as a function of the number of processes (1 and 2) and the number of generations (10, 15, and 20) using 24 threads per process. Fig. 3 shows the execution time obtained as a function of the number of threads (6, 12, and 24) and the number of generations (10, 15, and 20) using a single processor. The total memory use did not show dependence on the the number of threads. The memory use per process was rather dictated by the number of particles being simulated per process as well as the number of generations used with IFP. As expected, these results showed that the execution time improved significantly when more processes and more threads were used for the observed number of generations.

VI. CONCLUSIONS

The IFP method has been implemented, tested, and validated in OpenMC. This new feature allows users to compute effective kinetics parameters (effective delayed neutron fraction β_{eff} and effective generation time Λ_{eff}) for their OpenMC model by only adding a new setting and requesting three new tallies. It provides an easy way to generate the fundamental parameters needed to perform static and transient reactor safety analyses, and only necessitates one MC calculation as opposed to the two calculations involved in the perturbation-based approach. The new feature showed good agreement compared to computational results based on cases from the ICSBEP benchmark [13] reported for Serpent [2], MCNP [2], and a previous developmental branch of OpenMC [11] with

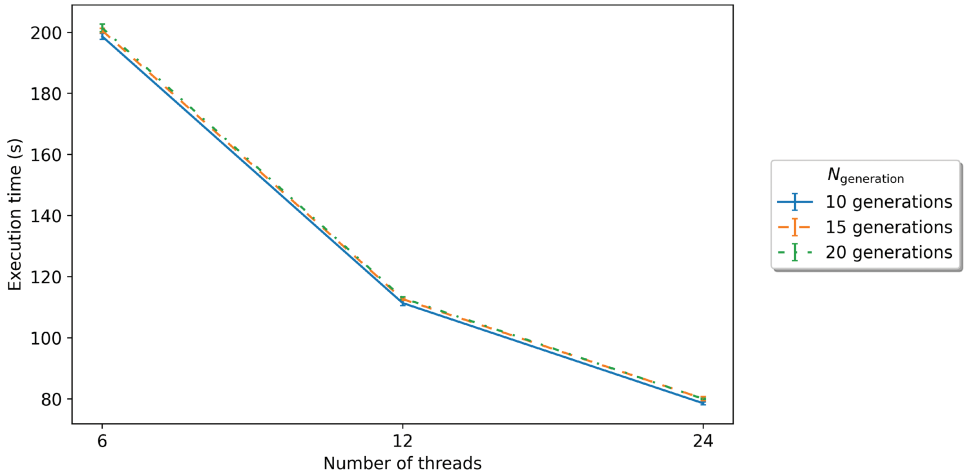


Fig. 3. Execution time as a function of the number of threads and IFP generations for single process calculations.

a maximum difference in β_{eff} of 25 pcm and 2.7% for Λ_{eff}. Comparisons to design data used in theoretical kinetics models for MSR experiments (ARE and MSRE) also showed reasonable agreement with a maximum difference in β_{eff} of 54 pcm and 13.4% for Λ_{eff}. The impact of the new implementation on code performance was verified to be minimal when IFP was not used with a maximum relative difference of 0.2% in peak memory usage and 0.8% in execution time for the model considered. When using the new IFP feature, the code performance scaled with the number of processes and threads, as expected.

Author Contributions

CRediT: **Joffrey Dorville:** Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Luke Labrie-Cleary:** Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Paul K. Romano:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing – review & editing.

Disclosure Statement

No potential conflict of interest was reported by the authors.

Data Availability Statement

The Python scripts used to generate the OpenMC models and OpenMC models in XML format for the ICSBEP benchmark are available at https://doi.org/10.5281/zenodo.17238768. All other data are available upon request.

Funding

This research was supported by a strategic partnership project with Copenhagen Atomics A/S. The material was based on work supported by the U.S. Department of Energy, Office of Science, under contract DE-AC02-06CH11357.

References

1. G. I. Bell and S. Glasstone, *Nuclear Reactor Theory*, Van Nostrand Reinhold Company, New York (1970).
2. J. Leppänen et al., "Calculation of Effective Point Kinetics Parameters in the Serpent 2 Monte Carlo Code," *Ann. Nucl. Energy*, **65**, 272 (2014); https://doi.org/10.1016/j.anucene.2013.10.032.
3. C. M. Perfetti, B. T. Rearden, and W. R. Martin, "SCALE Continuous-Energy Eigenvalue Sensitivity Coefficient Calculations," *Nucl. Sci. Eng.*, **182**, *3*, 332 (2016); https://doi.org/10.13182/NSE15-12.
4. C. M. Perfetti, "Advanced Monte Carlo Methods for Eigenvalue Sensitivity Coefficient Calculations," PhD Thesis, Nuclear Engineering and Radiological Sciences, The University of Michigan (2012).
5. B. C. Kiedrowski, F. B. Brown, and P. P. H. Wilson, "Adjoint-Weighted Tallies for k-Eigenvalue Calculations with Continuous-Energy Monte Carlo," *Nucl. Sci. Eng.*, **168**, *3*, 226 (2011); https://doi.org/10.13182/NSE10-22.
6. R. K. Meulekamp and S. C. van der Marck, "Calculating the Effective Delayed Neutron Fraction with Monte Carlo," *Nucl. Sci. Eng.*, **152**, *2*, 142 (2006); https://doi.org/10.13182/NSE03-107.
7. B. Verboomen, W. Haeck, and P. Baeten, "Monte Carlo Calculation of the Effective Neutron Generation Time," *Ann. Nucl. Energy*, **33**, *10*, 911 (2006); https://doi.org/10.1016/j.anucene.2006.05.001.
8. G. Truchet et al., "Computing Adjoint-Weighted Kinetics Parameters in Tripoli-4® by the Iterated Fission Probability Method," *Ann. Nucl. Energy*, **85**, *17*, 17 (2015); https://doi.org/10.1016/j.anucene.2015.04.025.
9. H. Hurwitz Jr., *Naval Reactors Physics Handbook*, Vol. 1, p. 864, A. Radkowsky, Ed., Naval Reactors, Division of Reactor Development, U.S. Atomic Energy Commission (1964).
10. Y. Nauchi and T. Kameyama, "Development of Calculation Technique for Iterated Fission Probability and Reactor Kinetic Parameters Using Continuous-Energy Monte Carlo Method," *J. Nucl. Sci. Technol.*, **47**, *11*, 977 (2010); https://doi.org/10.1080/18811248.2010.9711662.
11. X. Peng et al., "Calculation of Adjoint-Weighted Reactor Kinetics Parameters in OpenMC," *Ann. Nucl. Energy*, **128**, 231 (2019); https://doi.org/10.1016/j.anucene.2019.01.007.
12. P. K. Romano et al., "OpenMC: A State-of-the-Art Monte Carlo Code for Research and Development," *Ann. Nucl. Energy*, **82**, 90 (2015); https://doi.org/10.1016/j.anucene.2014.07.048.
13. *International Criticality Safety Benchmark Evaluation Project Handbook: ICSBEP Handbook 2021*, Organisation for Economic Co-operation and Development/Nuclear Energy Agency, Paris, France (2021); https://doi.org/10.1787/0235eb97-en.
14. A. Santamarina et al., "The JEFF-3.1.1 Nuclear Data Library —JEFF Report 22—Validation Results from JEF2.2 to JEFF-3.1.1," Organisation for Economic Co-operation and Development/Nuclear Energy Agency (2009).
15. M. Chadwick et al., "ENDF/B-VII.1 Nuclear Data for Science and Technology: Cross Sections, Covariances,

Fission Product Yields and Decay Data," *Nucl. Data Sheets*, **112**, *12*, 2887 (2011); https://doi.org/10.1016/j.nds.2011.11.002.

16. R. MacFarlane et al., "The NJOY Nuclear Data Processing System, Version 2016," LA-UR-17–20093, Los Alamos National Laboratory (2017).
17. S. J. Ball and T. W. Kerlin, "Stability Analysis of the Molten-Salt Reactor Experiment," ORNL-TM-1070, Oak Ridge National Laboratory (1965).
18. W. B. Cottrell et al., "Operation of the Aircraft Reactor Experiment," ORNL-1845, Oak Ridge National Laboratory (1955).
19. E. B. Knudsen and L. Chierici, "CAD_to_OpenMC: From CAD Design to Particle Transport," *J. Open Source Softw.*, **10**, *110*, 7710 (2025); https://doi.org/10.21105/joss.07710.
20. P. P. H. Wilson et al., "Acceleration Techniques for the Direct Use of CAD-Based Geometry in Fusion Neutronics Analysis," *Fusion Eng. Des.*, **85**, *10–12*, 1759 (2010); https://doi.org/10.1016/j.fusengdes.2010.05.030.
21. I. Wald et al., "Embree: A Kernel Framework for Efficient CPU Ray Tracing," *ACM Trans. Graph.*, **33**, *4*, 143:1 (2014); https://doi.org/10.1145/2601097.2601199.
22. P. Shriwise, " Double-Down," github; https://github.com/pshriwise/double-down.
23. T. J. Tautges et al., "MOAB: A Mesh-Oriented Database," SAND2004–1592, Sandia National Laboratories (2004); https://doi.org/10.2172/970174.
24. S. Yilmaz et al., "CAD and Constructive Solid Geometry Modeling of the Molten Salt Reactor Experiment with OpenMC," *Front. Nucl. Eng.*, **3** (2024); https://doi.org/10.3389/fnuen.2024.1385478.
25. D. Brown et al., "ENDF/B-VIII.0: The 8th Major Release of the Nuclear Reaction Data Library with CIELO-Project Cross Sections, New Standards and Thermal Scattering Data," *Nucl. Data Sheets*, **148**, *1*, 1 (2018); https://doi.org/10.1016/j.nds.2018.02.001.